MATH 128a: Homework #7

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Due by: **Wednesday, October 22, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 2.5

Solve problems 1d, 4, 6, 9, 15, 17 of Section 2.5 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 1

Exercise 2.5.1. The following sequences are linearly convergent. Generate the first five terms of the sequence $\langle \hat{p}_n \rangle$ using Aitken's Δ^2 method.

d. $p_0 = 0.5, \qquad p_n = \cos p_{n-1}, \qquad n \geq 1.$

Solution. Aitken's Δ^2 method has

$$\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}.$$

To get to the 5th term $\hat{p}_4$, we first need to compute p_1 through p_6. For this we have $p_0 = 0.5$, $p_1 = 0.877583$, $p_2 = 0.639012$, $p_3 = 0.802685$, $p_4 = 0.694778$, $p_5 = 0.768196$, and $p_6 = 0.719165$.

If we put these elements in an array, say A, and we define a function

$$g(x, y, z) = x - \frac{(y - x)^2}{z - 2y + x}$$

then we can systematically generate the terms of the Δ^2 sequence by calculating $g(A[i], A[i+1], A[i+2])$ for $i = 1 \ldots 5$. The result is: $\hat{p}_0 = 0.731385$, $\hat{p}_1 = 0.736087$, $\hat{p}_2 = 0.737653$, $\hat{p}_3 = 0.738469$, $\hat{p}_4 = 0.738798$

Note: if we were using Steffensen's method, we could just define the function

$$g(x) = x - \frac{(\cos x - x)^2}{\cos\cos x - 2\cos x + x}$$

and perform a basic fixed-point operation.

Exercise 2.5.4. Let $g(x) = 1 + (\sin x)^2$ and $p_0^{(0)} = 1$. Use Steffensen's method to find $p_0^{(1)}$ and $p_0^{(2)}$.

Solution. Let's implement Steffensen's method in a simple, rudimentary way without regard for ergonomics, numerical stability, or edge cases. The `diff` function is equivalent to the finite difference operator Δ. Note that MATLAB uses 1-based indexing.

```
format long;

f = @(x) 1 + (sin(x))^2;

p = [1, 0, 0];
p = iterate(f, p);
disp(p);

N = 5;
for i = 1:N
    p = aitken(p);
    p = iterate(f, p);
    disp(p);
end

function p = iterate(f, p)
    p(2) = f(p(1));
    p(3) = f(p(2));
end

function p = aitken(p)
    dp = diff(p);
    d2p = diff(dp);
    p(1) = p(1) - dp(1)^2/d2p(1);
end
```

Running this script gives you $p_0^{(k)}$ in the left-most column.

```
1.000000000000000   1.708073418273571   1.981273081117118
2.152904628999380   1.697735097378679   1.983972910564932
1.873464043514280   1.911155634772362   1.888560328658843
1.897028965802610   1.897294714091250   1.897133309671791
1.897194298133591   1.897194311322471   1.897194303311416
1.897194306338642   1.897194306338642   1.897194306338642
```

This these numbers are very close to the one reported by graphing calculators like Desmos. Exercise: These numbers are also close to $\pi/2$. You can use Taylor approximations to get the approximate solution

$$p \approx \frac{\pi - 1 + \sqrt{9 - 2\pi}}{2} \approx 1.89493.$$

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Exercise 2.5.6. Steffensen's method is applied to a function $g(x)$ using $p_0^{(0)} = 1$ and $p_1^{(0)} = \sqrt{2}$ to obtain $p_0^{(1)} = 2.7802$. What is $p_2^{(0)}$?

Solution. Steffensen's Method gives $p_0^{(1)} = p_0^{(0)} - \frac{(p_1^{(0)} - p_0^{(0)})^2}{p_2^{(0)} - 2p_1^{(0)} + p_0^{(0)}}$. So $2.7802 = 1 - \frac{(\sqrt{2}-1)^2}{p_2^{(0)} - 2\sqrt{2} + 1}$. So $p_2^{(0)} = 2\sqrt{2} - 1 - \left(\frac{(\sqrt{2}-1)^2}{2.7802 - 1}\right) \approx 1.73205$ (Notice that this is approximately $\sqrt{3}$.)

Exercise 2.5.9. Use Steffensen's method with $p_0 = 2$ to compute an approximation to $\sqrt{3}$ accurate to within 10^{-4}. Compare the results with those obtained in Exercise 11(9) of Section 2.2 and Exercise 14(12) of Section 2.1.

Solution. To use Steffensen's method, we first need a function for which $\sqrt{3}$ is a fixed point. In general, if you want to solve $f(x) = 0$ using a basic fixed-point iteration, a good function to try is $g(x) = x - f(x)/c$, where c has the same sign as $f'(p)$ and is large enough that $|g'(x)| \leq k < 1$ near $x = p$. With Steffensen's method, all you need is $g'(p) \neq 1$, but the point of the problem is to compare with the basic fixed-point iteration, so it's better to choose a $g(x)$ that works in that setting. Also, we avoid Newton's method (with $g(x) = x - f(x)/f'(x)$) since there is no need to accelerate convergence by Steffensen's method if $g'(p) = 0$ already. So a natural choice is $g(x) = x - (x^2 - 3)/3$, since $g'(x) = 1 - 2x/3$ is between $-1/3$ and $1/3$ for $x \in [1, 2]$. (Another idea that works well for this problem is $g(x) = \frac{x}{2} + \frac{3}{2x}$ on $[1.5, 2]$, except that it also converges quadratically, so does not need to be accelerated). For the calculation, we set

$$p_n = p_{n-1} - \frac{[g(p_{n-1}) - p_{n-1}]^2}{g(g(p_{n-1})) - 2g(p_{n-1}) + p_{n-1}},$$

and iterate starting with $p_0 = 2$. With $g(x) = x - (x^2 - 3)/3$, we get $p_1 = 1.727273$, $p_2 = 1.7320498$, $p_3 = 1.732050807569$. For comparison, Exercise 2.2.11(9) asks to solve the same problem by fixed-point iteration, while 2.1.14(12) asks to solve it via the bisection method. Fixed point iteration with $g(x) = x - (x^2 - 3)$ fails to converge since $|g'(\sqrt{3})| = |-2.46| \geq 1$. Using $g(x) = x - (x^2 - 3)/2$ works, but requires 51 iterations starting from $p_0 = 2$ to reach a tolerance of 10^{-7}. Using $g(x) = x - (x^2 - 3)/3$ works much better, converging to 10^{-7} in 10 iterations. The bisection method requires 24 iterations starting with $[a, b] = [1, 2]$ to converge to 10^{-7}. Steffensen's method is superior to both of these methods since it converges quadratically.

Exercise 2.5.15. Suppose that $\{p_n\}$ is superlinearly convergent to p. Show that

$$\lim_{n \to \infty} \frac{|p_{n+1} - p_n|}{|p_n - p|} = 1.$$

Solution. Recall that superlinear convergence means that the order of convergence α can be taken to be greater than 1. In other words, we have

$$\lim_{n \to \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = 0.$$

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There are three points at play here: p_n, p_{n+1}, and p. The intuition is that since p_{n+1} is much closer to p than p_n, the distance from p_n to p_{n+1} is essentially the same as the distance from p_n to p. Let's make this intuition rigorous. First, observe that

$$p_n - p = (p_n - p_{n+1}) + (p_{n+1} - p).$$

This equation expresses that the distance between p_n to p can be broken into two parts at p_{n+1}. Because of superlinear convergence, the first piece should be most of the distance, and the second piece should be a small minority of the distance. We can rearrange this equality to get

$$\frac{p_n - p_{n+1}}{p_n - p} = 1 - \frac{p_{n+1} - p}{p_n - p}.$$

All that remains is to take limits on both sides. Note that the absolute value of a limit is the limit of the absolute value, since the absolute value is a continuous function.

Exercise 2.5.17. Let $P_n(x)$ be the nth Taylor polynomial for $f(x) = e^x$ expanded about $x_0 = 0$.

a. For fixed x, show that $p_n = P_n(x)$ satisfies the hypotheses of Theorem 2.14.

b. Let $x = 1$, and use Aitken's Δ^2 method to generate the sequence $\hat{p}_0, \ldots, \hat{p}_8$.

c. Does Aitken's method accelerate convergence in this situation?

Solution. By Taylor's theorem, we have

$$f(x) = P_{n+1}(x) + R_{n+1}(x) = P_n(x) + \frac{f^{(n+1)}(0)}{(n+1)!}x^{n+1} + R_{n+1}(x).$$

Note that we expand by one additional term because we want to analyze the asymptotics of the error and not just upper bound it. Let $x^+ = \max\{x, 0\}$. Then we have

$$|R_{n+1}(x)| = \left|\frac{f^{(n+2)}(\xi_{n+1}(x))}{(n+2)!}x^{n+2}\right| \leq \exp(x^+)\left|\frac{x^{n+2}}{(n+2)!}\right|.$$

Note that $\xi_{n+1}(x)$ has a dependency on n, since it comes from an application of the mean value theorem to different derivatives of f of increasing order as $n \to \infty$. Now that we have attained a bound with a constant that does not depend on n, we have

$$f(x) - P_n(x) = \frac{x^{n+1}}{(n+1)!} + O\left(\frac{x^{n+2}}{(n+2)!}\right) = \frac{x^{n+1}}{(n+1)!}\left[1 + O\left(\frac{x}{n+2}\right)\right].$$

Be careful here, as the above Landau notation is for the limit as $n \to \infty$, not $x \to 0$. Remember that x is a fixed number. Observe that $O(x/(n+2)) = O(1/n) = o(1)$. This means that

$$f(x) - P_n(x) = \frac{x^{n+1}}{(n+1)!}(1 + o(1)).$$

It suffices to compute

$$\frac{f(x) - P_{n+1}(x)}{f(x) - P_n(x)} = \frac{x^{n+2}}{(n+2)!}(1 + o(1))\Big/\frac{x^{n+1}}{(n+1)!}(1 + o(1)) = \frac{x}{n+2}(1 + o(1)).$$

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Then $x/(n+2)$ vanishes as $n \to \infty$.

It is also possible to do this problem without the additional term in Taylor's theorem. You just have to include a lower bound for the error. Simply write

$$\exp(x^-) \leq \exp(\xi_n(x)) \leq \exp(x^+),$$

where $x^- = \min\{x, 0\}$. Then

$$R_n(x) = \frac{f^{(n+1)}(\xi_n(x))}{(n+1)!}x^{n+1} = \frac{\exp(\xi_n(x))}{(n+1)!}x^{n+1}$$

and thus

$$\left|\frac{R_{n+1}(x)}{R_n(x)}\right| = \left|\frac{\exp(\xi_{n+1}(x) - \xi_n(x))}{n+2}x\right| \leq \left|\frac{xe^{|x|}}{n+2}\right|.$$

Again, this vanishes as $n \to \infty$.

Note that the above actually shows that the convergence is superlinear. That said, $P_n(x) \to f(x)$ does not converge with order strictly greater than 1. It is possible to extract the critical exponent by taking

$$\alpha = \lim_{n\to\infty} \frac{\log R_{n+1}(x)}{\log R_n(x)}.$$

It can be shown that $\log \Theta(\beta_n) = \log \beta_n + O(1)$ and that $\log \beta_n(1 + o(1)) = \log \beta_n + o(1)$. Either of these can be used alongside Stirling's approximation to show that the critical exponent is 1. In general, it is possible to have superlinear convergence without converging with order higher than 1. This just means that you have a bound $\epsilon_{n+1} \leq g(\epsilon_n)$ with g a linear function, and that you can make this bound sharper by replacing g not with a stronger power law, but some other function, often involving logarithms. In this case, the sharp bound involves the Lambert W function.

To show that Aitken's Δ^2 process still improves the convergence in the superlinear case, we can analyze the expression for the improvement ratio in the proof of Theorem 2.14.

$$\frac{\hat{p}_n - p}{p_n - p} = 1 - \frac{1}{\epsilon_n} \cdot \frac{(\Delta\epsilon_n)^2}{\Delta^2\epsilon_n}.$$

Reminder that this comes from the calculation from class:

$$\hat{\epsilon}_n = \epsilon_n - \frac{(\Delta\epsilon_n)^2}{\Delta^2\epsilon_n}.$$

In our case, $p_n = P_n(x)$, $p = f(x)$, and $\epsilon_n = R_n(x)$. So long as $\epsilon_{n+1} = o(\epsilon_n)$, we have

$$\Delta\epsilon_n = \epsilon_n(1 + o(1)) \text{ and } \Delta^2\epsilon_n = \epsilon_n(1 + o(1)).$$

Then the improvement ratio remains 0 as in the linear case. That said, in the superlinear case, the convergence of the original sequence will already be very fast, and it's possible that Aitken's method won't be an improvement over simply going out an additional few terms, especially given that Aitken's Δ^2 process requires that you compute extra terms anyways. This is why we often say

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that Aitken's method offers no practical benefit when $\alpha > 1$. You should actually be comparing $\hat{\epsilon}_n$ to ϵ_{n+2}, and it can be shown that the latter is better when $\alpha > 1$. Consider why comparing $\hat{\epsilon}_n$ to ϵ_n is sufficient when $\alpha = 1$.

We can compute Aitken's Δ^2 process with the following script:

```
N = 8;
n = 0:N;
p = cumsum(1./factorial(n));
dp = 1./factorial(n + 1);
d2p = -1./factorial(n)./(n + 2);
disp([p; dp; d2p; p - dp.^2./d2p]');
```

We use closed-form expressions for the finite differences to avoid catastrophic cancellation. Running this yields:

```
    1.0000    1.0000   -0.5000    3.0000
    2.0000    0.5000   -0.3333    2.7500
    2.5000    0.1667   -0.1250    2.7222
    2.6667    0.0417   -0.0333    2.7188
    2.7083    0.0083   -0.0069    2.7183
    2.7167    0.0014   -0.0012    2.7183
    2.7181    0.0002   -0.0002    2.7183
    2.7183    0.0000   -0.0000    2.7183
    2.7183    0.0000   -0.0000    2.7183
```

Aitken's Δ^2 process gives 5 correct digits more quickly than the original process. This suggests that the convergence is faster. This is also what we expect from Theorem 2.14.

Problem 2: Newton's method

The function

$$f(x) = \frac{x}{\sqrt{1+x^2}}$$

has a unique root at $x = 0$.

a) Show that Newton's method gives $x_{k+1} = -x_k^3$ (where x_k is the approximation to the root obtained in the kth iteration and x_{k+1} is the approximation obtained in the $k+1$th iteration.). Explain why the method succeeds in approximating the root if and only if $|x_0| < 1$.

b) Draw graphs to illustrate the first few iterates when $x_0 = 0.5$ and when $x_0 = 1.5$.

Answer of Problem 2

We compute the derivative

$$f'(x) = \frac{1}{(1+x^2)^{3/2}}.$$

This means that

$$x - \frac{f(x)}{f'(x)} = x - \frac{x}{\sqrt{1+x^2}}(1+x^2)^{3/2} = x - x(1+x^2) = -x^3.$$

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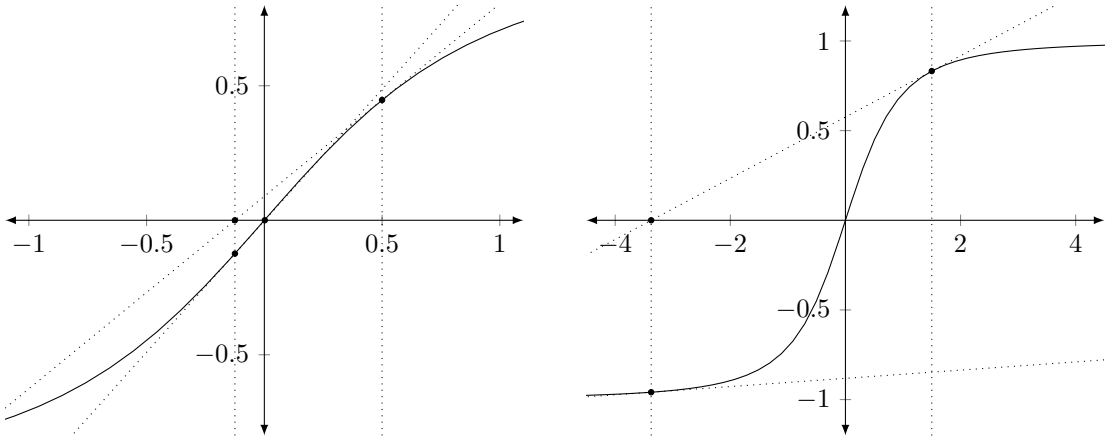
If x_k denotes the sequence produced by Newton's method, we have

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = -x_k^3.$$

This means that Newton's method on $f(x)$ is equivalent to fixed-point iteration on $g(x) = -x^3$. Indeed, g has a unique fixed point at $x = 0$. We can analyze the convergence directly without taking derivatives or considering the contraction factor. We start with x_0, and each application of g both reverses the sign and multiplies the power by 3. One can use induction to show that

$$x_k = (-1)^k x_0^{3^k} \text{ and thus } \lim_{k\to\infty} |x_k| = \lim_{k\to\infty} |x_0|^{3^k} = \lim_{n\to\infty} |x_0|^n = \begin{cases} 0 & |x_0| < 1 \\ 1 & |x_0| = 1 \\ +\infty & |x_0| > 1 \end{cases}.$$

Below are plots of Newton's method for $x_0 = 0.5$ and $x_0 = 1.5$.



Problem 3: Stopping criterion for fixed point iteration

Let $\Phi(x) = x$ a fixed point equation with a fixed point x^* which is not explicitly known, and a sequence x_k, $k \geq 1$ computed using the nonlinear fixed point iteration from some starting value x_0. Assume that the sequence x_k converges to x^* as $k \to \infty$ and that $|\Phi'(x)| \leq q < 1$ for all $x \in \mathbb{R}$. If you would like to get an approximation x_K for some $K \in \mathbb{N}$ to the root x^* that satisfies $|x^* - x^K| < \tau$ for some given tolerance $\tau > 0$, what is a good stopping criterion for the iteration? Explain your choice with rigorous arguments. The expression should NOT depend on x^*.

Answer of Problem 3

Because we want to target an absolute error tolerance, it makes sense to use some sort of absolute difference for our stopping criterion. If you have an explicit universal bound q on the derivative, the relationship between $|x_{K-1} - x_K|$ and $|x_K - x^*|$ is known and explicit. Namely, telescoping gives us

$$\sum_{k=0}^{\infty} (x_{K+k} - x_{K+k+1}) = x_K - \lim_{k\to\infty} x_{K+k+1} = x_K - x^*.$$

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Additionally, induction on $|\Delta x_{K+k+1}| \leq q|\Delta x_{K+k}|$ gives us $|\Delta x_{K+k}| \leq q^k|\Delta x_K| \leq q^{k+1}|\Delta x_{K-1}|$. Then an application of triangle inequality followed by the explicit formula for geometric series yields

$$|x_K - x^*| \leq \sum_{k=0}^{\infty} |x_{K+k} - x_{K+k+1}| \leq \sum_{k=0}^{\infty} q^{k+1}|x_{K-1} - x_K| = \frac{q}{1-q}|x_{K-1} - x_K|.$$

Then to ensure that $|x_K - x^*| < \tau$, it suffices that

$$|x_{K-1} - x_K| < \left(\frac{1}{q} - 1\right)\tau.$$

Notice that this criterion can be checked with knowledge of just q, τ, and the computed sequence.

It is possible to take the first difference and use a logarithm base q to determine how many steps it takes to reach the tolerance. However, this could be an overestimate, since Φ could contract more than q for some of the intervening applications.

When you compute K iterations, the information you will have seen consists of all the points x_k for $k \leq K$. It is best to use the most recent data for your estimate.

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